

LAPORAN PRAKTIKUM JARINGAN KOMPUTER WEEK 14

Modul 14 802.11 WiFi

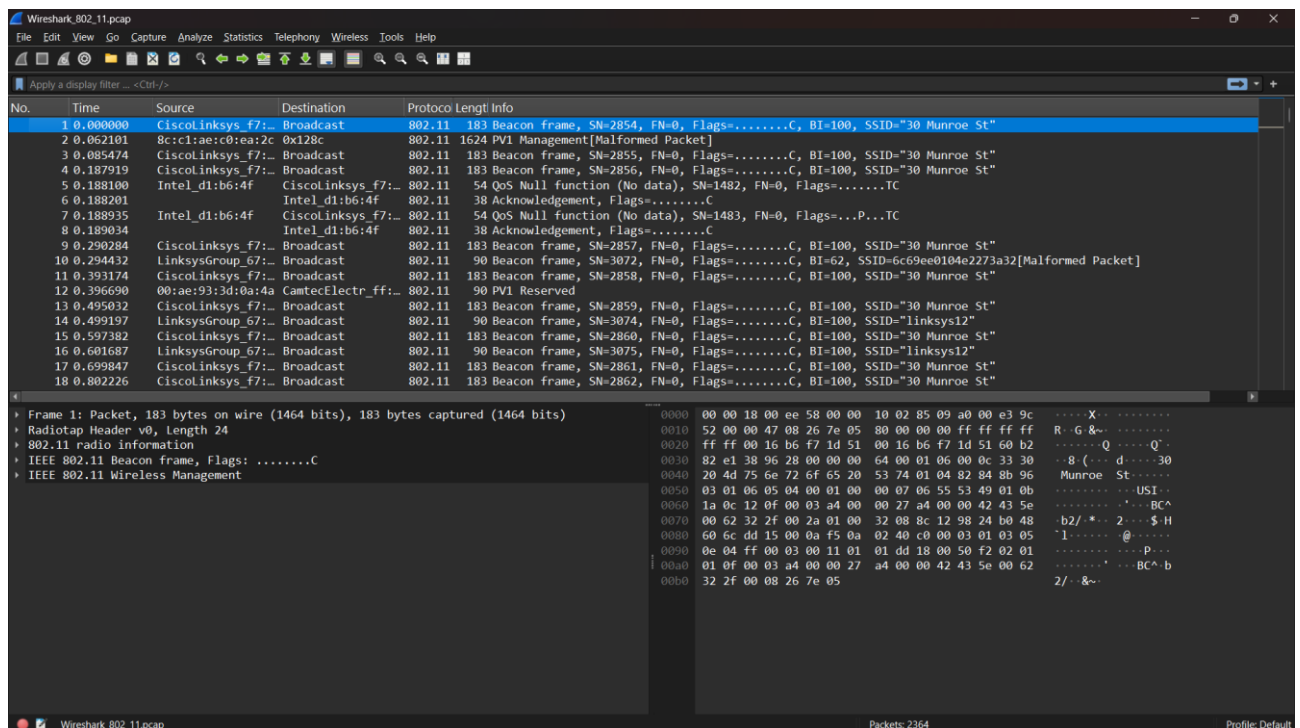
Tujuan Praktikum :

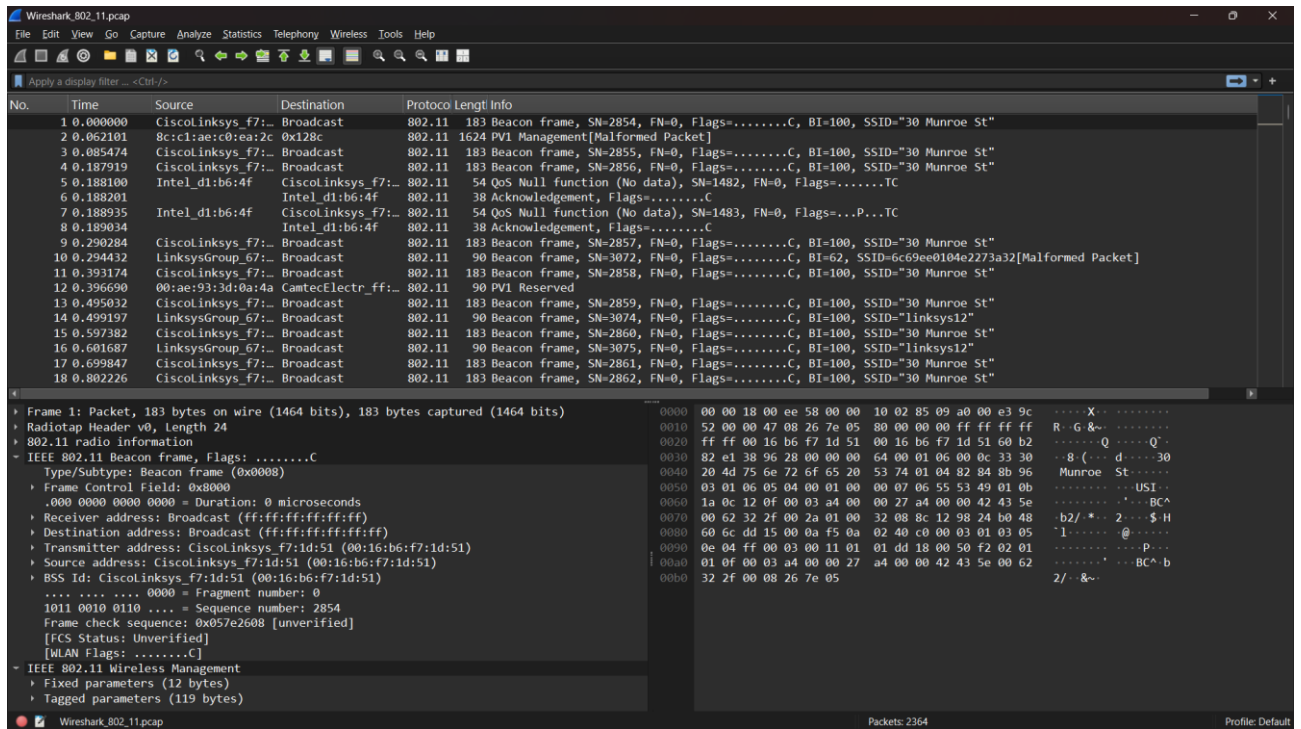
1. Cara kerja protokol WiFi IEEE 802.11
2. Struktur frame 802.11
3. Beacon Frame
4. Data Transfer pada jaringan WiFi
5. Association dan Disassociation antara client dan Access Point (AP) menggunakan Wireshark.

WireShark_802_11.pcap

Aktivitas host nirkabel yang diambil dalam file jejak adalah:

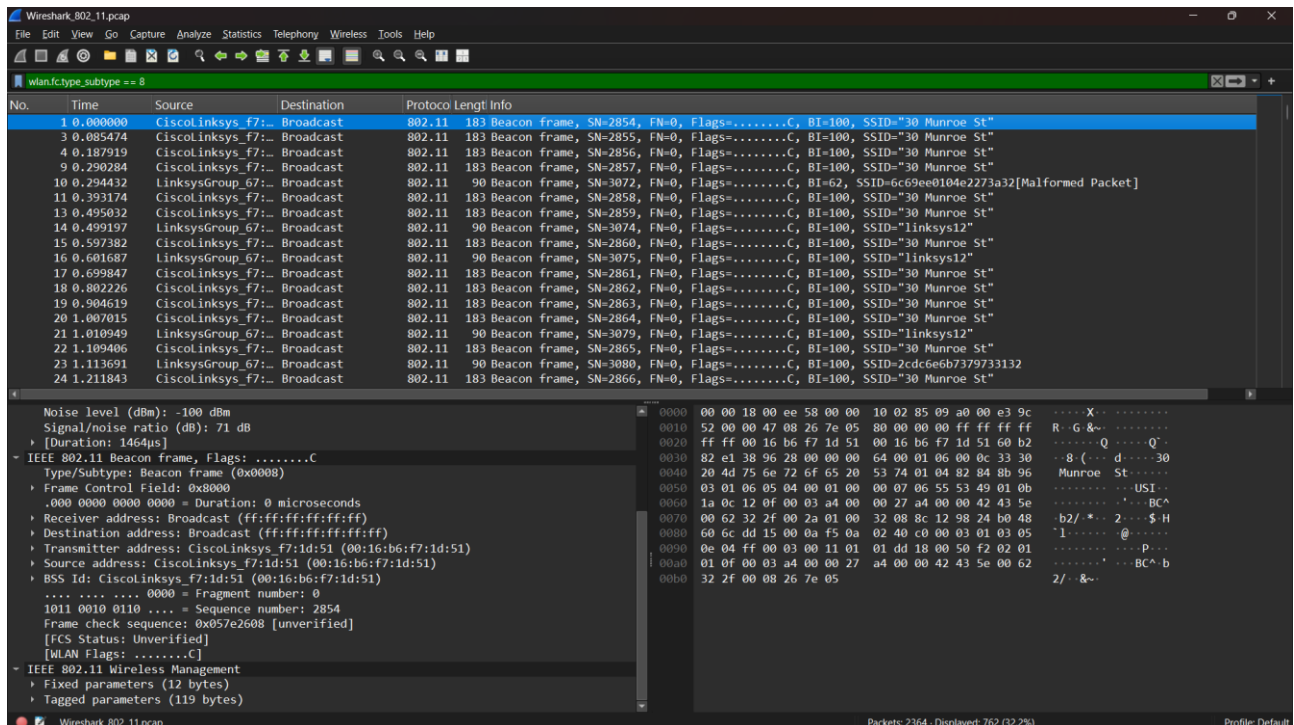
1. Host sudah diasosiasikan dengan 30 Munroe St AP saat pelacakan dimulai. Pada t = 24,82, host membuat permintaan HTTP ke <http://gaia.cs.umass.edu/wiresharklabs/alice.txt>. Alamat IP gaia.cs.umass.edu adalah 128.119.245.12.
2. Pada t=32.82, host membuat permintaan HTTP ke <http://www.cs.umass.edu>, dengan alamat IP 128.119.240.19. Pada t = 49,58, host memutuskan sambungan dari 30 Munroe St AP dan mencoba menyambung ke linksys_ses_24086. Ini bukan titik akses terbuka, dan host akhirnya tidak dapat terhubung ke AP ini.
3. Pada t=63.0 tuan rumah menyerahkan mencoba untuk mengasosiasikan dengan linksys_ses_24086 AP, dan mengasosiasikan lagi dengan titik akses 30 Munroe St.





Beacon Frames

Fungsi : Mengumumkan keberadaan jaringan WiFi (SSID) dan memberikan informasi konfigurasi jaringan kepada perangkat di sekitar.



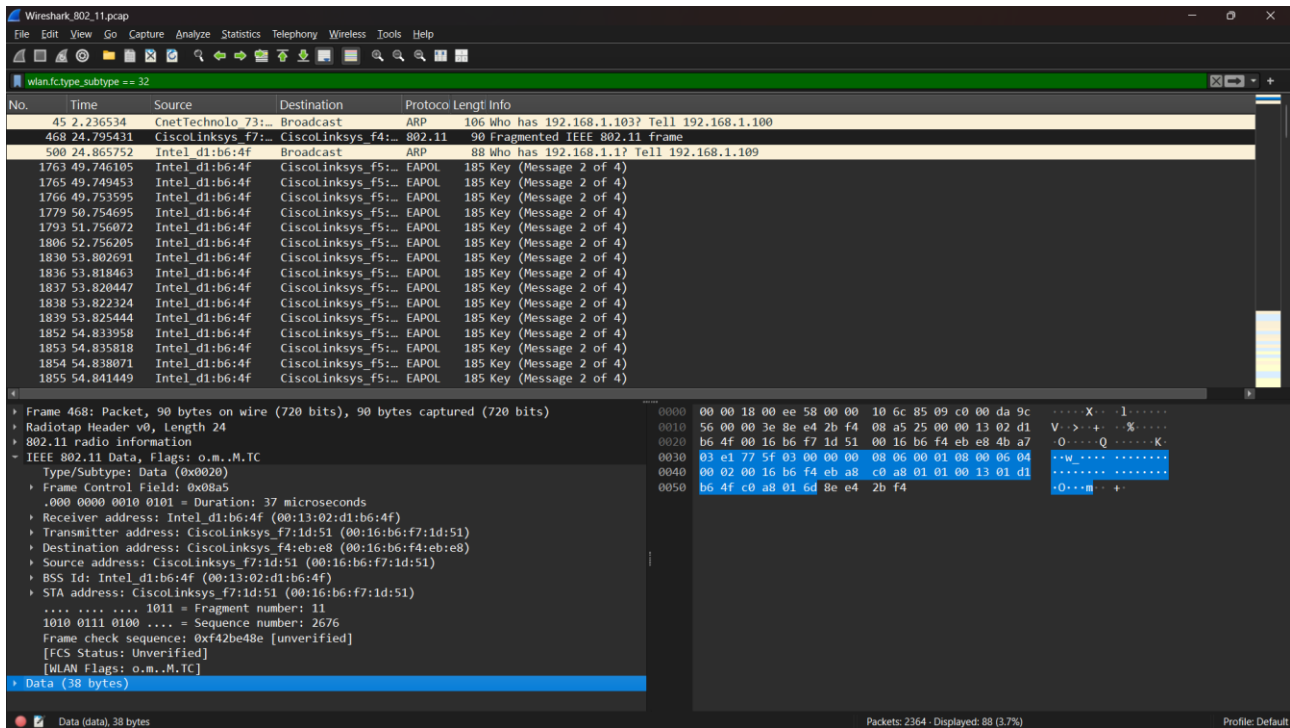
Analisis :

1. SSID = 30 Munroe St
2. BSSID = 00:16:b6:f7:1d:51

3. SOURCEADDRESS = 00:16:b6:f7:1d:51
4. DESTINATIONADDRESS = ff:ff:ff:ff:ff:ff
5. Beacon Interval = BI = 100 Beacon Interval = 100 TU (Time Unit)

Data Transfer

Fungsi : Mengirimkan data pengguna seperti HTTP, TCP, DNS, dan aplikasi lainnya melalui jaringan WiFi.



Association/Disassociation

Fungsi Association Request : Association Request adalah frame yang dikirim oleh client (station) kepada Access Point untuk meminta izin bergabung ke jaringan WiFi.

The image shows a Wireshark packet capture of IEEE 802.11 Association Request frames. The top pane displays a list of captured packets, with packet 1933 selected. The middle pane shows the details of the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
1227	33.079714	d1:b6:4f:00:16:b6	MS-NLB-PhysServe...	802.11	111	Association Request, SN=3775, FN=4, Flags=.pm...F.C
1750	49.651078	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1607, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1751	49.653218	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1607, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1824	53.789944	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1613, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1825	53.790943	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1613, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1827	53.793568	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1613, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1926	57.903699	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1927	57.904945	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1932	57.911195	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1933	57.915945	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1934	57.924199	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1935	57.936216	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1937	57.939196	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
2126	62.176945	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1645, FN=0, Flags=.....C, SSID="linksys_SES_24086"
2127	62.178194	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	107	Association Request, SN=1645, FN=0, Flags=.....C, SSID="linksys_SES_24086"
2162	63.169910	Intel_d1:b6:4f	CiscoLinksys_f5:...	802.11	89	Association Request, SN=1648, FN=0, Flags=.....C, SSID="30 Munroe St"
2307	70.179949	CiscoLinksys_f5:...	f9:ff:ff:ff:ff:ff	802.11	132	Fragmented IEEE 802.11 frame

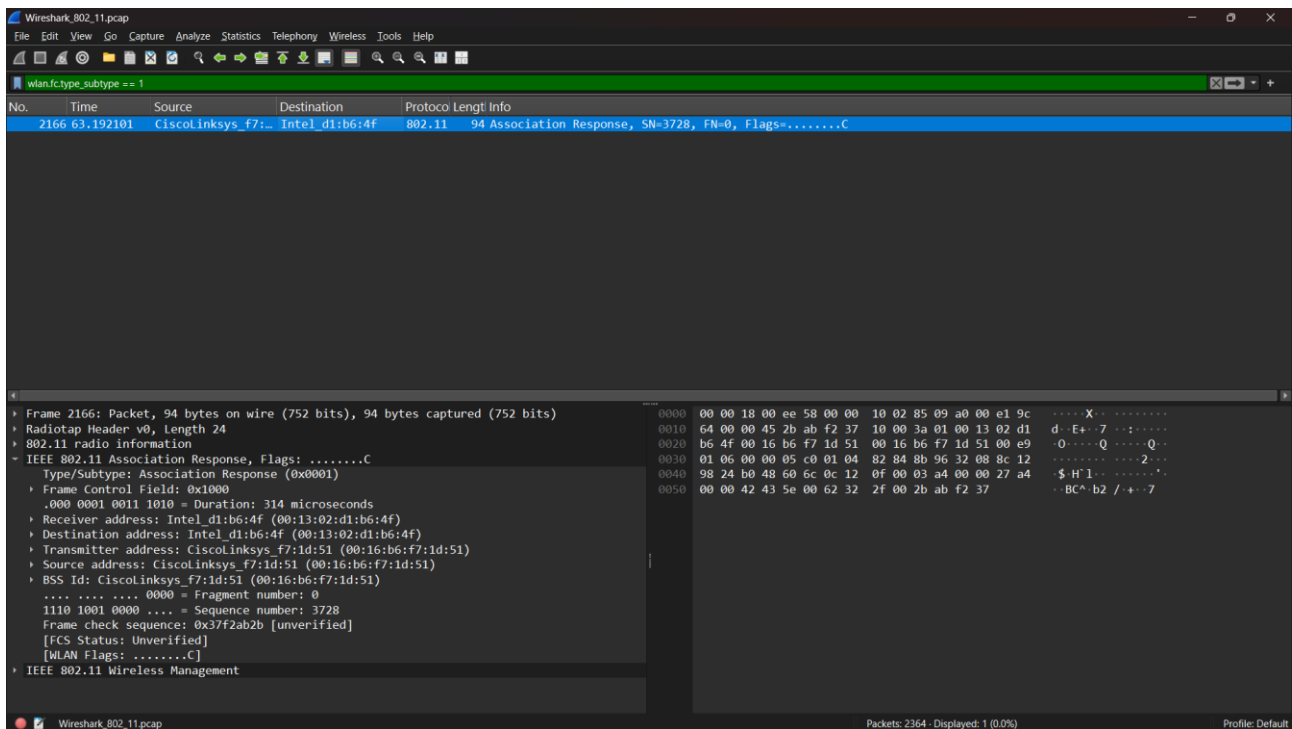
Packet Details (Packet 1933):

- Frame 1227: Packet, 111 bytes on wire (888 bits), 111 bytes captured (888 bits)
- Radiotap Header v0, Length 24
- 802.11 radio information
- IEEE 802.11 Association Request, Flags: .pm...F.C
 - Type/Subtype: Association Request (0x0000)
 - Frame Control Field: 0x0062
 - Duration/ID: 2048 (reserved)
 - Receiver address: MS-NLB-PhysServer-32_08:00:00:13:02 (02:28:00:00:13:02)
 - Destination address: MS-NLB-PhysServer-32_08:00:00:13:02 (02:28:00:00:13:02)
 - Transmitter address: d1:b6:4f:00:16:b6 (d1:b6:4f:00:16:b6)
 - Source address: d1:b6:4f:00:16:b6 (d1:b6:4f:00:16:b6)
 - BSS Id: f7:1d:51:00:16:b6 (f7:1d:51:00:16:b6)
 - STA address: MS-NLB-PhysServer-32_08:00:00:13:02 (02:28:00:00:13:02)
 - 0100 = Fragment number: 4
 - 1110 1011 1111 = Sequence number: 3775
 - Frame check sequence: 0x775edca5 [unverified]
 - [FCS Status: Unverified]
 - [WLAN Flags: .pm...F.C]
 - WEP parameters
- Data (51 bytes)

Raw Data (Hex):

```
0000 00 00 18 00 ee 58 00 00 10 00 85 09 c0 00 dd 9c .....X.....
0010 00 00 00 41 a5 dc 5e 77 00 02 00 88 02 28 00 00 ...A...w...b...
0020 13 02 d1 b6 4f 00 16 b6 f7 1d 51 00 16 b6 f4 eb ...O...Q.....
0030 a8 00 d3 00 00 aa aa 03 00 00 00 08 00 45 20 00 .....E.....
0040 28 ca 0e 00 00 29 06 09 3b 40 e9 bb 68 c0 a8 01 (...)...;@-h...
0050 6d 00 50 09 f2 aa de 4d 70 fc 84 7a 6f 50 10 19 m-P...M p...zoP...
0060 20 5e c8 00 00 00 00 00 00 00 a5 dc 5e 77 A.....w
```

Fungsi Association Response : Balasan dari Access Point terhadap permintaan Association Request dari client.



Fungsi Disassociation : Disassociation digunakan untuk mengakhiri hubungan antara client (station) dan Access Point pada jaringan WiFi. Tidak ditemukan frame Disassociation pada file capture Wireshark_802_11.pcap yang digunakan dalam praktikum. Hal ini menunjukkan bahwa selama proses perekaman tidak terjadi pemutusan hubungan antara client dan Access Point, atau proses tersebut tidak ikut terekam dalam capture.